

Virendra Bais

Solution Architect With GenAI /Data Engineering Specialization On Python/Java Platforms

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Innovative and hands-on **genAI Architect** and **Data Engineering** expert with enterprise-level Architect Certifications (TOGAF 9.1 and SCEA 5) and a proven track record in designing cutting-edge **AI** and **Agentic AI** solutions.

Specialized in **Generative AI, Machine Learning**, and cloud-based architectures, particularly on Google Cloud Platform (GCP). Extensive hands-on experience in developing end-to-end **GenAI applications**, exemplified by a personal project that provides one-click AI-based analysis of company quarterly results.

Currently serving as Vice President, Technology at Deutsche Bank, with a rich background in leading financial institutions and tech giants. Demonstrated ability to architect and implement business-critical **AI systems**, including advanced **analytics platforms** and AI-driven corporate banking solutions. Adept at collaborating with C-level executives to align AI and data engineering strategies with business objectives, driving innovation, and optimizing large-scale enterprise solutions.

Skills

AI/ML Architecture: GenAI, LLMs, AI Agents,, RAG (Retrieval Augmented Generation)

Enterprise Architecture: TOGAF, Solution Design, Technology Strategy

Google Cloud Platform (GCP), Kubernetes, Docker

Data Engineering: Big Data, Apache Spark, Hadoop, Hive, Apache Airflow

Programming: Python, Java, LAMP stack

- **DevOps & CI/CD:** Jenkins, Maven, Git, JIRA
- Application Security & Scalability

Technology Experience

AI/ML: Extensive experience with Generative AI, Large Language Models (LLMs), and Retrieval Augmented Generation (RAG) and Agentic applications

Big Data: Spark, Hadoop, Hive, Hue, Apache Airflow etc.

Google Cloud: BigQuery, Dataproc, Dataflow, PubSub etc. Dataflow

Java/JEE: Spring, TensorFlow, Keras, Pandas, NumPy, PyTorch, Drupal

DevOps: OpenShift, Kubernetes, Docker, Jenkins, Maven, JIRA, TeamCity

Databases: Milvus Vector Database, MongoDB, Oracle, MySQL, MS SQL, DB2

Work History

2016-04 - Current

Vice President, Technology

Deutsche Bank, Pune, India

- **Architecting and building GenAI based systems:**

Some notable projects:

- Graph RAG based specification creator for automotive functional safety
- Flex to Angular migration using GenAI and Agents
- Domain Aware Text Querying DB using RAG
- Confluence research agent
- GenAI driven insights into company financials.
- **Contributor to open source GenAI RAG framework llmware** (llmware.ai) which enables end-to-end GenAI project implementation using RAG Architecture and Agents. (Can be verified from the project founders).
- **Led innovation and AI adoption** chapter in the Credit Risk department, generating 40+ innovative ideas and shepherding 15+ prototypes to production, with 5+ currently in active use.
- Led architectural transformation of the Credit Risk Stress Testing platform using **Hadoop** and **Spark**, resulting in a 66% performance improvement and enhanced scalability. Designed and implemented a core system component with Databricks-like functionalities, replacing a costly SAAS reporting tool and saving \$1 Million
- Architected and built a bank-wide platform for static scanning and remediation of security vulnerabilities.

2012-04 - 2016-04

Senior Architect, TAO (Technology Architecture Office)

Cognizant Technologies

Senior Architect – Organization-wide Architecture Horizontal

Directed architectural strategy and decision-making for the department, collaborating with key stakeholders, including clients, project managers,

developers, and technical leads to ensure alignment with business objectives and enterprise standards.

Lead Architect – RBS, UK

Provided architectural leadership for a department of 170+ professionals, overseeing the design and direction of new applications. Ensured that all solutions were scalable and innovative, aligned with the bank's strategic goals, and supported long-term digital transformation objectives. Led the re-architecture of the bank's corporate banking portal from a monolithic architecture to a service-oriented architecture (SOA), enhancing flexibility, modularity, and scalability

1998-08 - 2012-04

Associate Consultant (Enterprise Architect)

Tata Consultancy Services

Enterprise Architect – Capital Markets & Financial Services, TCS Delivered strategic architecture consultancy and solution architecture across the Capital Markets and Financial Services horizontal group, guiding architecture consultation, design reviews and security evaluations for development teams and clients. Oversaw critical project deliveries to ensure alignment with business and technical goals.

Lead Architect – Online Brokerage Platform, CIBC Canada

Led the re-architecture and design of the front-end for CIBC's Online Brokerage site, a critical platform for one of Canada's 'Big Five' banks. Oversaw the modernization of a platform comprising 500 screens, processing 5 million transactions per day, and supporting 6,000 concurrent users. This initiative significantly enhanced brand value and positioned the business at the forefront of digital capabilities.

Solution Owner & Architect – TCS iON (SMB Solutions)

Led the solutioning, architecture, and technical implementation of 3 SaaS based solutions for TCS iON, targeted at Small and Medium Businesses (SMB). Delivered scalable, multi-tenant architecture solutions that supported TCS iON's go-to-market strategy and drove innovation in the SMB sector.

Tech Lead – ShipServ.com

Architected and developed the initial version of ShipServ.com, a pioneering B2B SaaS marketplace for the shipping industry, connecting global ship operators with suppliers. This solution laid the foundation for a billion dollar business, transforming procurement and supply chain operations across the maritime sector.

Certifications

TOGAF 9.1

Oracle Certified Enterprise Architect (SCEA 5)

Certified SAFe (Scaled Agile Framework) Practitioner

Education

1994-07 - 1998-05

Bachelor of Engineering: Electronics And Communications Engineering

National Institute of Technology, Jaipur

Graduated with **Honours** in Electronics and Communications Engineering